

A photograph of a wind farm in a hilly landscape under a blue sky with white clouds. Several white wind turbines are visible, with one in the foreground being the most prominent. The hills are covered in green grass and some trees.

U.S. WIND FARM ACQUISITIONS

Strategic Data Analysis

Exploratory Data Analysis Report

Introduction & Background

Wind energy is booming in the United States. As of 2021, there were **over 70,000 wind turbines** across 46 states, generating **140 GW of power**.

For a private equity firm, simply buying any wind farm isn't enough. They need to understand the nuances: which models are most efficient, how technology has evolved, and which regions offer the best wind conditions.

70,808

Total Turbines in Dataset

140+ GW

Installed U.S. Capacity

46

States with Turbines

27

Data Variables Analyzed

Problem Statement & Project Goals

THE CHALLENGE

Our firm wants to acquire U.S. wind farms, but we need to know **what drives peak energy efficiency** . We must understand what a historically successful wind farm looks like before investing.

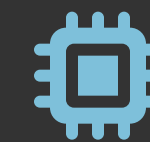
THE GOAL

Identify the best geographical regions, turbine specifications, and wind conditions to guide our investments and **maximize ROI** .



Geographic Focus

Find states with most high-capacity turbines



Technology Track

Track how turbine specs have improved over the decades



Wind Resource

Alignment

Combine turbine data with external wind speed and electricity cost data to find the best market opportunities

Data & Methodology

We analyzed **70,808 turbine records** and merged with state-level wind speed and electricity pricing datasets.



U.S. Wind Turbine Database

70,808 turbine records from Lawrence Berkeley National Laboratory with 27 variables



Wind Speed Data

State-level average wind speeds across all 50 U.S. states



Electricity Pricing

State-level residential and commercial electricity rates

ANALYSIS PIPELINE



The EDA Process

We analyzed **70,808 turbine records** and This is the steps we took in the process.

Auditing & Cleaning

Evaluated missing data (e.g., identifying that 91% of turbines have never been retrofitted) and fixed formatting errors.

1

Feature Engineering

Mapped states to U.S. Census Regions and grouped exact installation years into 10-year "Decade Buckets" for trend tracking.

2

Data Aggregation

Rolled up 70,000 individual turbine records into state-level summaries to calculate total counts and average megawatt capacities.

3

External Integration

Merged state-level wind speeds with our turbine data, categorizing continuous wind speeds into distinct "Wind Buckets" (Low, Moderate, High).

4

Agenda

Our roadmap for this strategic analysis

01

Geographic Concentration

Mapping turbine distribution across
states and Census Regions



02

Technology Evolution

Tracking capacity, hub height, and
rotor diameter improvements over
time



03

Wind Resource Alignment

Cross-referencing wind speeds with
turbine density and capacity



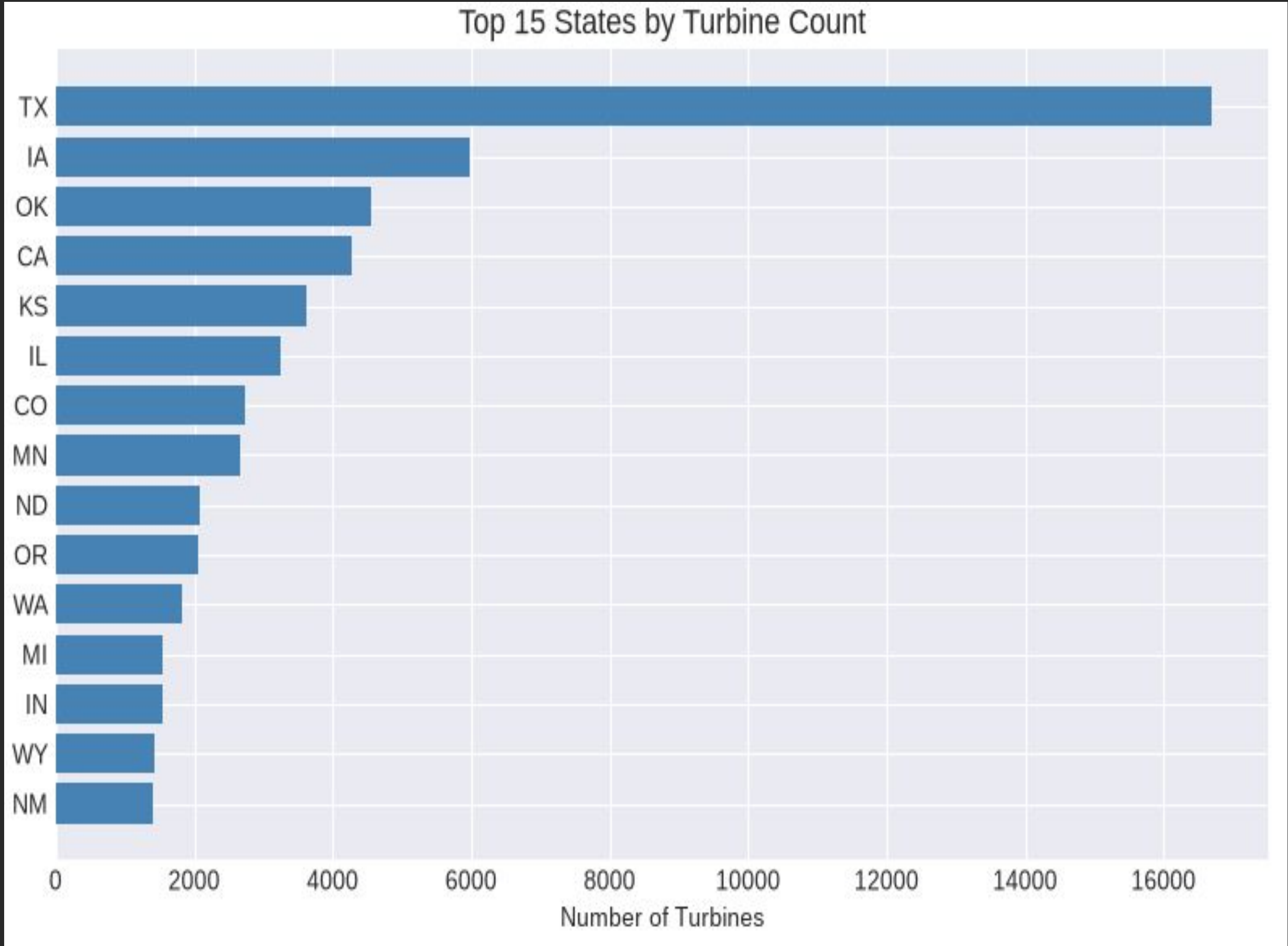
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Recommendations

Actionable investment strategies for
the acquisition committee

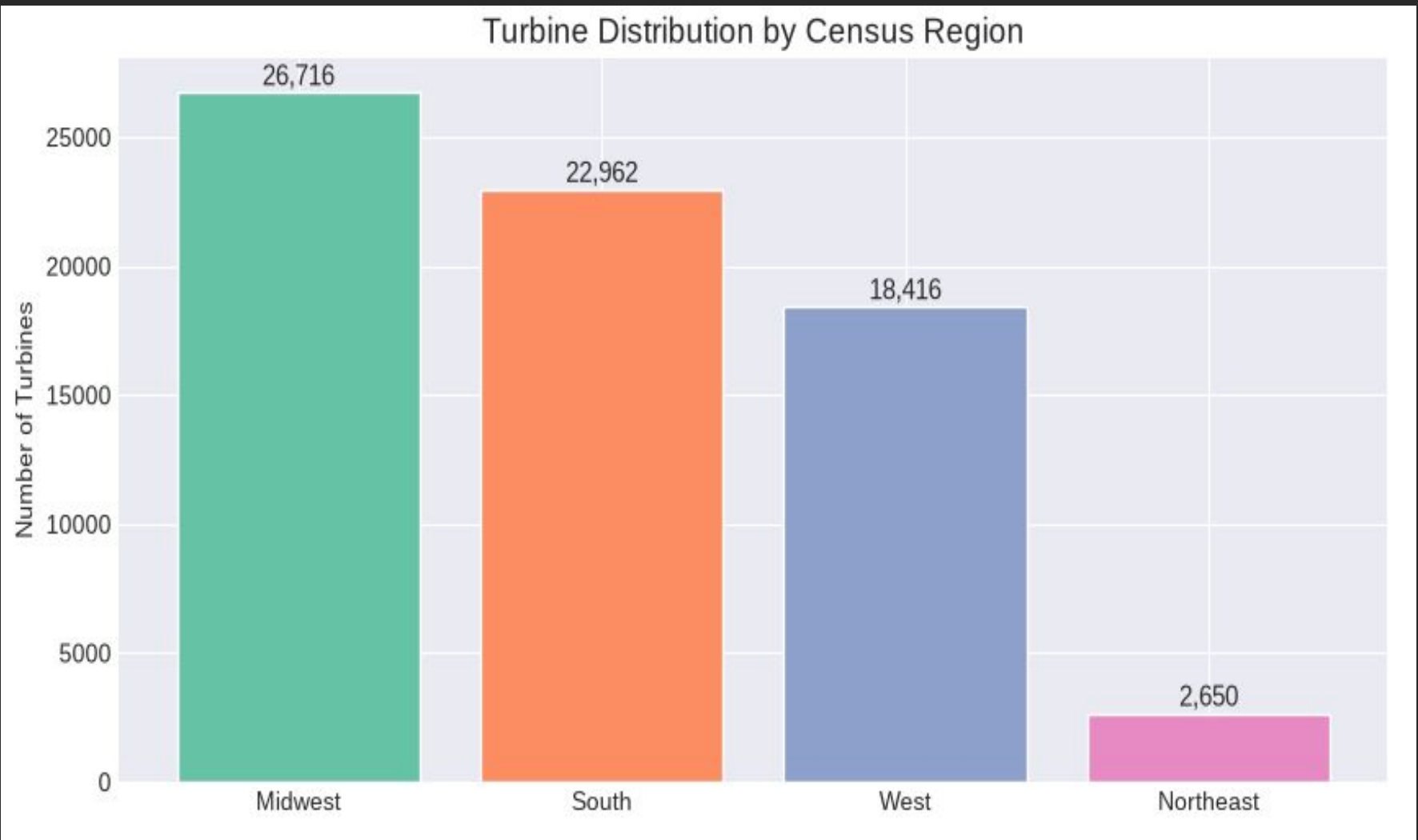


Geographic Dominance



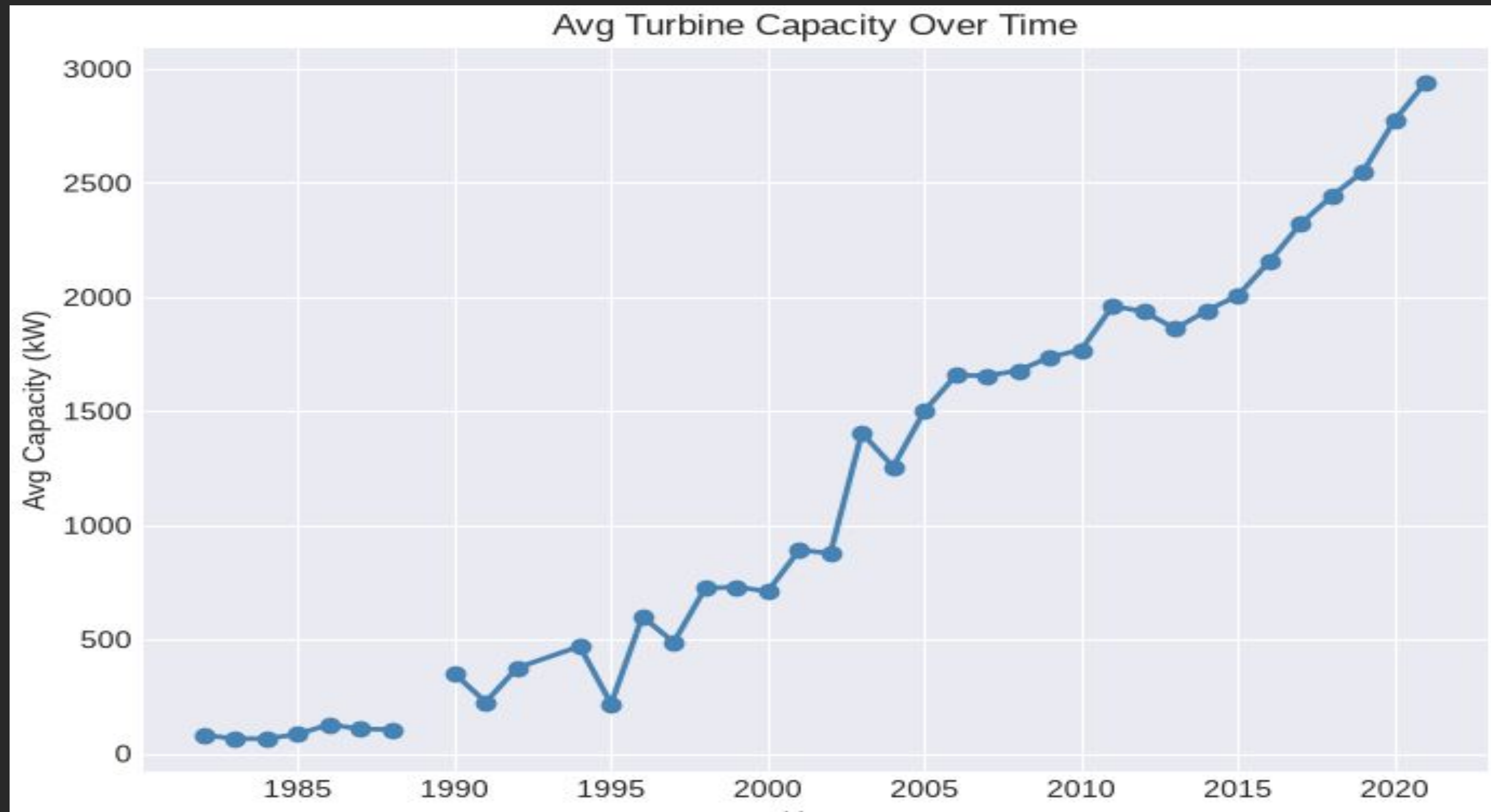
KEY INSIGHTS

- **Texas dominates** with 16,688 turbines — more than the entire Western U.S. combined
- **Midwest leads regions** with ~26,700 turbines (38.5% of all U.S. installations)
- **Top 5 states** (TX, IA, OK, CA, KS) hold 55% of all U.S. turbines



Technology Evolution

Average turbine capacity has grown 5x since the 1990s



Key Takeaway: If we are buying older farms, we are buying outdated tech. Newer is significantly better — post-2015 turbines average >2,300 kW.

~500

kW avg in 1990s

~3,000

kW avg by 2020

90m+

Modern hub height (vs. 50m in 1990s)

130m+

Modern rotor diameter (vs. 50m in 1990s)

The Wind "Sweet Spot"

Cross-referencing wind speeds with turbine installations

Wind Bucket	Turbine Count	Avg Capacity (kW)	Avg Hub Height (m)	Avg Rotor Diam (m)
Low (<14 mph)	4,290	1,361	63.1	66.8
Moderate (14-16)	20,188	2,037	81.3	97.9
Good (16-18)	13,143	2,053	83.1	98.2
High (18-20)	21,026	1,962	82.9	97.1
Very High (>20)	6,618	1,957	80.7	96.4

KEY INSIGHTS

- Highest wind states (>20 mph: SD, MT, WY) are **underbuilt** — only 6,618 turbines
- The **sweet spot** is "High" wind (18-20 mph) with 21,026 turbines — optimal density + resource alignment
- Low-wind states (<14 mph) show **significantly lower capacity** — avoid for acquisitions

Summary of Findings

01

Location is Highly Concentrated

Texas and the Midwest dominate the market with over 43% of all U.S. turbines.

Texas alone has 16,688 turbines.

02

Size Matters Enormously

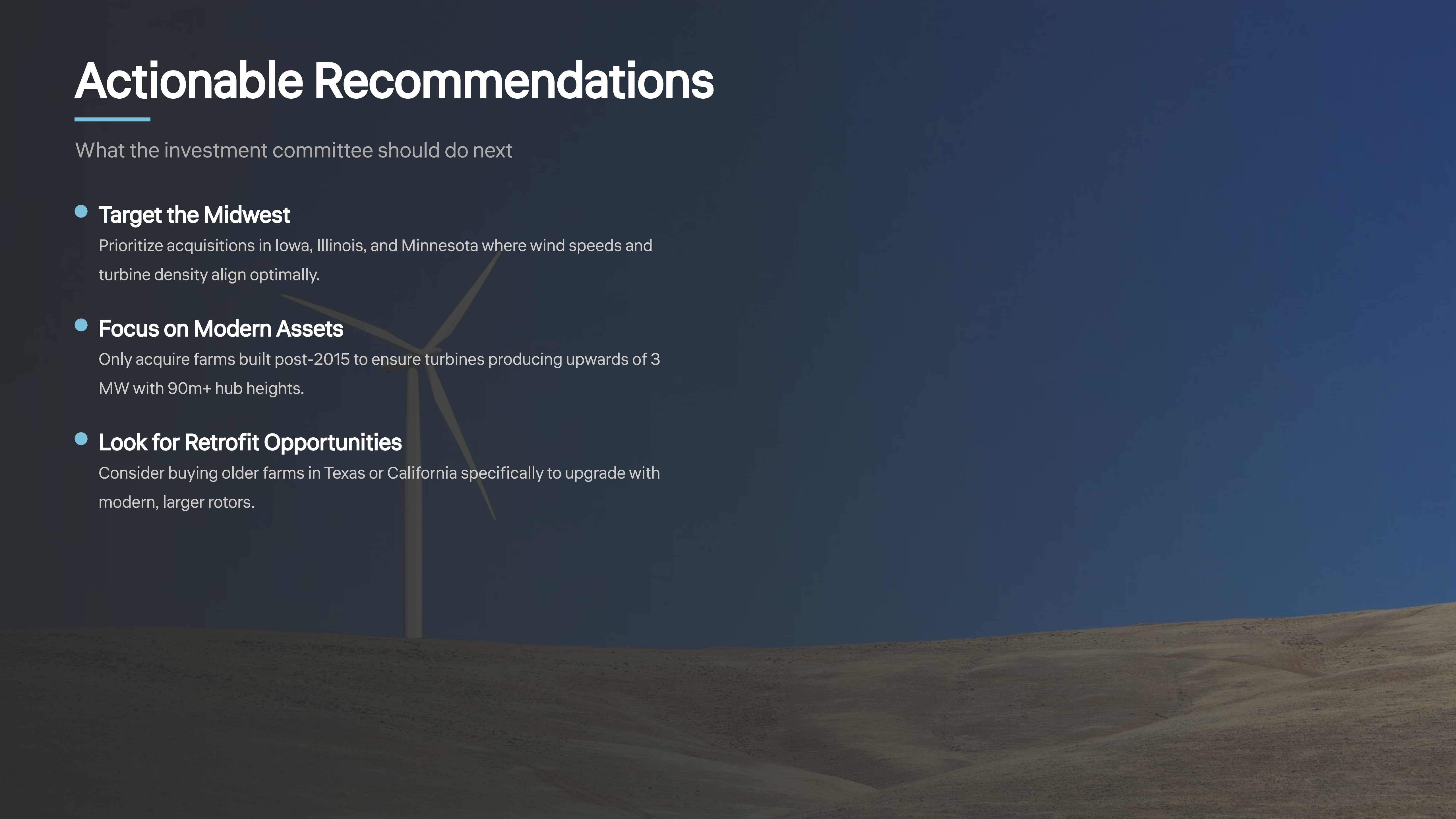
Post-2015 turbines average >2,300 kW with 90m+ hub heights — exponentially more powerful than pre-2005 models.

03

Untapped Potential Exists

High-wind states (SD, MT, WY >20 mph) are underbuilt. The sweet spot is 18-20 mph range with optimal density.

Actionable Recommendations



What the investment committee should do next

- **Target the Midwest**

Prioritize acquisitions in Iowa, Illinois, and Minnesota where wind speeds and turbine density align optimally.

- **Focus on Modern Assets**

Only acquire farms built post-2015 to ensure turbines producing upwards of 3 MW with 90m+ hub heights.

- **Look for Retrofit Opportunities**

Consider buying older farms in Texas or California specifically to upgrade with modern, larger rotors.